

Formal Verification: 6CCS3VER

Exercise Sheet 7

Important Note: Please download Isabelle from the link provided in the slides and bring a laptop to the large group tutorial. Performing the proofs in Isabelle will be an integral part of those tutorials.

Before you start, please open a new Isabelle/HOL theory file called `Week_1.thy` and add the following to its beginning.

```
theory Week_2
imports Main
begin
```

Exercise 7.1 Addition Commutes

Consider the function $(+)$ defined in Isabelle/HOL for natural numbers.

Prove the following theorem, first pen-and-paper, and then formally in Isabelle.

lemma *add_commutes*: “ $(n::nat) + m = m + n$ ”

Hints: 1. Perform the proof by induction on n , using substitutions *subst*, and forward reasoning using the method *rule*.

2. You will need to identify multiple lemmas about how *add* works. Do not prove those lemmas. Use *sorry* as the proof. This is the top-down approach.

3. There is a proof which uses the following lemmas:

add_0_right, *add_0_left*, *add_Suc_right*, *add_Suc*

Exercise 7.2 Multiplication is Monotone

Consider the multiplication function $(*)$ defined in Isabelle. Prove the following property of it, i.e. that it is monotonically increasing in both arguments. You should prove it both, pen-and-paper and in Isabelle.

lemma *mult_mono*:

fixes $a\ b\ c\ d::nat$ — Note the alternative way of fixing the types of the constants

assumes “ $a \leq b$ ” “ $c \leq d$ ”

shows “ $a * c \leq b * d$ ”

Your pen-and-paper proof should indicate

1. what lemmas are you assuming,
2. on what variable are you performing induction,
3. what are assumptions and the proof goals in the base case and the step case, and what is the induction hypothesis, and
4. how are you instantiating the induction hypothesis, i.e. how do you show that its assumptions hold and which quantified variables are instantiated with which constants.

Hints: 1. The proof should be by induction

2. The following lemmas suffice to prove the goal with only substitution, forward reasoning, and induction:

refl, *diff_le_self*, *mult_le_mono2*, *mult_zero_right*, *le0*, *mult_Suc_right*, *add_mono*, *Suc_eq_plus1*, *le_diff_conv*

3. When you have a term like $x - y$, where x is a *nat*, in the goal or the assumptions, you should perform proof by case analysis on whether $y \leq x$. This is because of the way $-$ is defined for natural numbers, where $x - y = 0$, for any $x \leq y$.